

An Authentication Mechanism Based on Zero Trust With Radio Frequency Fingerprint for Internet of Things Networks

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Abstract—With the development of Internet of Things (IoT) and cloud networks, the security of edge networks, borderless networks, and obscure networks are essential, so there are many security problems that need to be tackled, including over-trust in trust areas and security only based on security boundaries in traditional security architecture. According to characters of zero trust security architectures and integrative trust model (ITM), the zero trust architectures (ZTAs) better adapt to handle these security problems compared to the ITM for the IoT networks. Meanwhile, the radio frequency fingerprint (RFF) identification keeps high accuracy and high stability with researchers' investigation, which makes RFF authentication feasible. Therefore, we propose a mechanism that combines the RFF authentication technique and ZTA to improve security in IOT networks, including edge networks, borderless networks, and obscure networks. The method resolves the difficulty of over-reliance on a trustable center or trust chain, and the method is suitable for borderless networks and obscure networks. Besides, this method resists data leakage, counterfeit attack and rouge access point (AP) attack with RFF authentication, and it can reduce the risk caused by compromised devices with zero trust concepts. With the analysis in this article, the proposed method keeps high-level security and performance that method effectively against spoofing identity, tampering, and information disclosure. The authentication accuracy of the method has reached 99%, and the authentication owns robustness in time cost and collision-resistant.

Index Terms—Authentication, cyber security, Internet of Things (IoT), IoT security, radio frequency fingerprint (RFF), zero trust.

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I. INTRODUCTION

THE DEVELOPMENT and deployment of the Internet of Things (IoT) and cloud networks make bordered networks turn into obscure or borderless networks, and various network applications and network services make network become more complex compared with the traditional network, so a novel method to handle these problems is important to be proposed. It is obvious that the consequences of malicious attacks seriously damage assets and disrupt real society [1]. For example, in 2022, Colonial Pipeline, a top fuel passage operator in the United States, was attacked with a ransomware attack, making fuel supply failure within 17 states of the USA for three days. As illustrated in Fig. 1, the IoT networks rely on wireless network to take communication between gateway and terminal devices, and there are three inherent problems to break network system security. Meanwhile, the development of zero trust and physical security is increasing, so a method distinguished from the traditional security approach that zero trust combined with radio frequency fingerprint (RFF) authentication is a feasible approach. Zero trust architectures (ZTAs) own the advantages for dealing with borderless networks and obscure networks, and ZTAs increase data security in storage and transmission. Besides, RFF authentication offers more secure authentication on the access point (AP), and it offers complex and various authentications combined with traditional authentications, reducing risk with different authenticators and different authentication methods. The method using zero trust and RFF demonstrates a certain degree of security and performance compared with traditional architecture and methods. However, this mechanism cannot support long-range authentication, because it requires an authenticator to receive the signals from the client.

For instance, in the vehicular networks scenario, the system architecture includes a root trusted authority (TA), numerous distributed roadside units (RSUs), and vehicles. The root TA is responsible for recovering the identity of the illegal vehicle, and the RSUs are responsible for distributing secrets [2]. Therefore, there are high-level security devices, TAs, and low-level security devices, RSUs, in this scenario, so there is vulnerable to the compromise of TA due to relying on a trustable center. Besides, the communications between RSUs and vehicles are vulnerable to rouge AP attacks, spoofing identity, tampering, information disclosure, etc. However, our

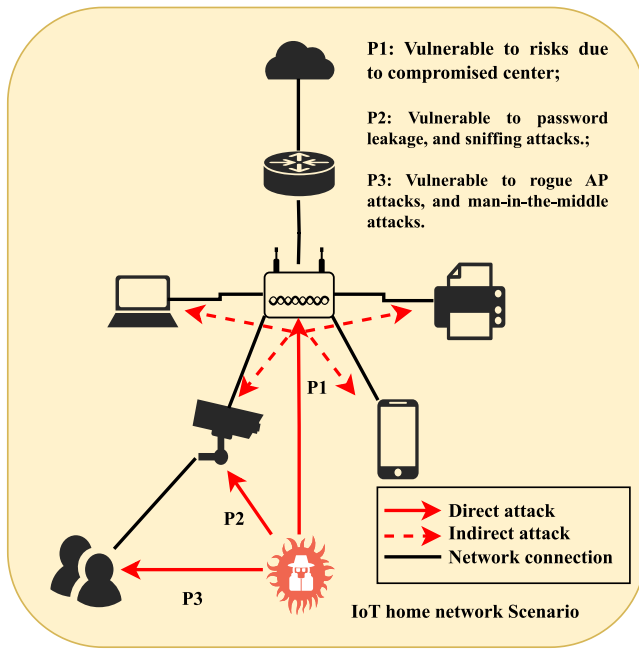


Fig. 1. Problems about wireless network in the IoT scenario.

proposed mechanism using multiple authenticators and authentication methods with RFF can ensure the removal of the center. In addition, we utilize RFF authentication to handle the risk of authentications between users and low-level security authenticators, which eliminates rogue AP attacks, spoofing identity, tampering, and information disclosure effectively based on RFF features, such as no-cloning of RFF.

Secure authentication is a major difficulty in the cyber security field. Previous researches and practical applications are mostly based on the integrative trust model (ITM) and password authentication, so investigation on zero trust and RFF authentication is significant. The reason why investigate zero trust and RFF authentication is that giving a mechanism combining RFF authentication and zero trust to overcome the drawbacks of traditional architecture and password authentication.

The traditional architecture means the ITM, it owns a credible center to keep other elements secure within the system, so the center is the weakness of this model, including high operating costs and delay, and risk of system and data [3]. If the center is compromised, the security of this system will be broken, which is an inherent issue. Nevertheless, zero trust will reduce the terrible influences if the center is compromised, because ZTA removes the dependence on the security center and reduce risk with constant control, constant monitoring, and continuous multielement feedback. Therefore, zero trust could be used to overcome overreliance on security centers in traditional architecture.

The IoT currently requires secure technologies in authentication methods, and the mechanism we designed is suitable for IoT wireless network scenarios, because the RFF authentication is based on the wireless network and it keeps high security with the features of RFF [4]. In addition, considering

the characteristics of IoT that limited energy and computation power [5], it is reasonable to implement RFF authentication to handle most authentication with the low cost of energy and computation power [6], meanwhile, the sensitive information of password will seldom transfer over the IoT network, which reduces the risk of credentials exposure.

Besides, password authentication owns disadvantages that attacks are straightforward to operate, and attack automation is uncomplicated. For example, password eavesdropping, brute force attack, social engineering, and side channel attack. This kind of problem frequently crops up that simple and default password is easy to be cracked and used by users [7], complex password often has been written down, and multiple accounts use the same password for an user. In addition, rogue AP attack still is difficulty for password authentication in the wireless network. Therefore, RFF authentication is necessary to be used in the network, because it is able to overcome these shortages of password authentication based on analysis in Sections III and IV. Meanwhile, there is a gap in applying RFF authentication within ZTA, the RFF authentication ensures placing authentication on an AP or APs is secure and possible, which could instead of duplicated verification within a traditional architecture.

In this article, we investigate a method based on zero trust with RFF, which makes contributions to secure authentication studies with the proposed method. This study suggests a methodological approach with zero trust conception and compares it with the classical authentication method. The contributions are listed as follows.

- 1) The proposed method takes zero trust security architecture, which is suitable for IoT networks, including borderless networks and obscure networks compared with the traditional security architecture, and this mechanism is able to secure IoT devices and networks. The ways taken include decentralization, application of RFF authentication and password authentication, security policies, and multiple authenticators, which offer constant control and risk allocation following the conceptions of zero trust and reduce the risk of password disclosure.
- 2) We evaluate the proposed mechanism in terms of security and performance with experiments and threat model. In the security analysis, we find that the proposed method resists information disclosure well, including rogue AP attacks, password leakage, and sniffing attacks with STRIDE model. In the performance evaluation, we find that the proposed mechanism keeps high accuracy and high robustness with the experiments' results. With these evaluations, the combination between zero trust and RFF authentication is feasible in the practical scenario, and it is promising in the further study.
- 3) We set up an experimental platform to test the security and performance of the proposed mechanism. With the experimental platform, we are able to find the differences between RFF authentication and password authentication in security and performance, including accuracy experiments, time cost experiments, and collision experiments, which highlight the usability and

security of RFF techniques for securing devices at the access layer.

Following the introduction below, Section II gives a background and related work about IoT wireless security problems, zero trust, and radio frequency fingerprinting technique. Section III presents the method to authenticate the wireless IoT device with ZTA and RFF techniques. Section IV elaborates the security analysis about the method proposed. Section V evaluates the performance of the proposed method. Section VI summarizes this article.

II. BACKGROUND AND RELATED WORK

This section includes IoT wireless security problems, zero trust, and radio frequency fingerprinting technique. We first point out the IoT wireless security problems within the traditional security architecture, then we introduce the zero trust is able to handle these problems with its characters. Besides, RFF technique can support the implementation of zero trust and secure access layer with its authentication theory and method.

A. IoT Wireless Security Problems

IoT owns great development and deployment prospect with various IoT devices, but the security of IoT becomes a new problem. For example, BASHLITE in 2014, Mirai [8] in 2016, and Hajime in 2017.

The unique features of IoT devices make the IoT vulnerable, including user-centric, Internet-connected, and complex [9]. User-centric means the IoT devices not only store the users' sensitive information and small IoT devices do not enable privacy-preserving algorithms [10] but also the IoT devices could make damage to users. Internet-connected means IoT devices are able to directly or indirectly connect to the Internet, which suffers a large-scale attack surface. Complex means IoT devices and IoT systems become more sophisticated, so the number of vulnerabilities will increase due to it is a positive correlation with systematic complexity, and the traditional full data encryption cannot apply to complex IoT scenarios [11]. Meanwhile, the authentication is related to these characteristics, the user-centric makes the credentials exposure, and the Internet-connected and complex make authentication vulnerabilities. Therefore, using RFF authentication to supplement security combined password authentication is a great choice, which is presented in Section III.

Moreover, the limitations of IoT devices are computational power and energy, so it contributes to vulnerabilities that insufficient energy harvesting, inadequate authentication, improper encryption, insufficient access control, and insufficient audit mechanisms. According to the study from [12], effect on energy consumption increases 230 % when IoT device enables three security functions.

A common attack that access attack means hackers access to network or device without authorization. For example, exploitation of weak or nonexistent passwords, and exploitation of services like HTTP, FTP, and SNMP. Besides, rogue AP attack is a common attack on wireless devices, most users will access the wireless device with the strongest signal, then if a

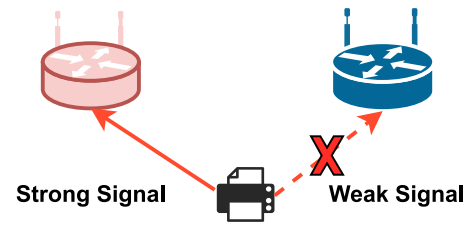


Fig. 2. Rogue AP attack.

rogue AP owns a strong signal, the user will access the rogue AP like Fig. 2. Then, the traffic and sensitive information, such as passwords, would be achieved by the attacker setting this rogue AP. Nevertheless, our mechanism deals with these problems in Section III.

Costin et al. [13] found that returning the password hashes to the plaintext passwords are feasible by experiments. The author takes a large-scale static analysis on firmware, and 55% of retrieved password hashes are recovered successfully. However, the RFF authentication our mechanism used against this attack with the passwordless feature to secure IoT devices.

In addition, [14] claims that there are some protocols that IoT devices used are vulnerable. For example, AMQP protocol is vulnerable to authentication bypass due to improper encryption on user credentials. Therefore, the proper proposed mechanism is designed in Section III and RFF authentication can avoid threats caused by password authentication.

Considering the needs of IoT devices in wireless networks that low bandwidth, low computation power, and low energy, [15] believes that RFF is a cost-effective security solution. Meanwhile, the proposed authentication mechanism does get not rid of password authentication, we utilize the RFF authentication to recover the drawbacks of the classical password-based authentication mechanism.

B. Zero Trust

The zero trust concept originated from Jericho Forum in 2004 for offering a new architecture and solution in the borderless network. The term, “zero trust,” is proposed formally by John Kindervag in 2016 to provide a cyber security solution to handle network security concerns that security systems are overly reliant on trust and trusted systems [16]. The need for zero trust is noticeable, because the network boundary shows the trend of becoming borderless or blurred due to the technological development of cloud and IoT. Our method utilizes zero trust concept to supplement traditional security architecture for reducing risk and adapting to borderless and obscure networks.

The main idea of zero trust is “never trust, always verify,” it means each element inside or outside the network should not be trusted by default, and the authentication and authorization mechanism should be careful, namely, authorization or authentication repeatedly instead of long-term valid authorization or authentication. Besides, zero trust could require credible credentials, including multifactors that IP address, MAC address, user id, password, geographic location, face, eyes, etc. In addition, the ZTA should rely on the removal of traditional

TABLE I
COMPARISON OF KEY ASSUMPTIONS AND ELEMENTS DRAWN FROM ITM WITH THOSE HELD IN (ZT/ZTA) [17]

Element	Comparison of key assumptions	
	Trust: Integrative trust model (ITM)	Zero trust and zero trust architectures (ZT/ZTA)
Start-point(assumption)	Positive expectations of trust	Expectations of threats, or malicious actors (inside or out)
Assessment	Ability, benevolence, integrity	Threat traffic: authentication and authorization
Propensity	Multidimensional	No presumptive trust
Vulnerability	Willingness to be vulnerable	Not willing to accept vulnerability
Uncertainty	Suspension of uncertainty	Uncertainty is a given aim to minimize
Risk	Risk taking	Not willing to take/ aim to reduce risk
Outcomes	Trust	Authentication and authorization: trust
Control	Absence of control	Constant control
Monitoring	Absence of monitoring	Constant monitoring
Feedback loop	End point	Continuous multielement feedback
Level	Individual, team, organization	Subject: user (human), non-human (service, device, application)

inherent trust networks, because the authority is generated by the authorization or authentication from the center. Therefore, we present a solution for problems of security centralization in Section III-A removal of center with our mechanism.

Wylde et al. [17] suggested a table that comparison of the key assumptions and elements drawn from the ITM with those in (ZT/ZTA). According to Table I given, there are many aspects existing differences. The characters should be highlighted from ZTAs include the following points.

In terms of start-point (assumption), the ITM holds a positive attitude toward the trust chain or trust architecture, but the ZTA assumes malicious actors exist inside or outside the system. In terms of control, ITM lacks control, because it is always based on the trust for a security center offering authentication and authorization, but ZTA should offer constant control, namely, multiple security mechanisms. Therefore, we utilize RFF technique to support the implementation of zero trust and improve security in Section IV-B RFF authentication for improving security. In terms of risk, ITM takes risk due to admitting of willingness to be vulnerable, but the ZTA does not want to take risk and wants to reduce the risk. Therefore, we present various designs in Section IV to decrease systemic risk.

In addition to Table I, [18] suggests a comparison between zero trust security architecture and traditional architecture. The protection objects in traditional security architecture focus on the “network,” which brings problems that cannot handle borderless networks or obscure boundary networks efficiently, such as cloud networks. However, zero trust security architecture focuses on the flexible boundary and no trust, which can adapt to obscure or borderless networks.

In summary, the advantages of zero trust security architecture are dealing with insider attacks and handling borderless networks or obscure boundary networks by architecture. According to the rapid development of cloud networks and

IoT, the borderless network and obscure boundary network scenarios increase, so ZTA owns advantages compared with traditional security architecture in ability and security. Meanwhile, the zero trust needs a secure authentication method to support it, so we utilize RFF authentication to ensure secure authentication at the access layer with its unique characters.

C. Radio Frequency Fingerprinting Technique

The conception of physical-layer device identification has been proposed for defensive and offensive purposes. Toonstra and Kinsner [19], Choe et al. [20], and Hall et al. [21] raised physical-layer devices identification that could be used for intrusion detection; [22] and [23] discuss access control with physical-layer devices identification; [24] considers worm-hole detection with physical-layer devices identification; [25] and [26] think about cloning detection with physical-layer devices identification; [27] discusses malfunction detection with physical-layer devices identification; [28] argues about secure localization; and [29] thrashes out rogue AP detection with physical-layer devices identification. Before [30] proposes RFF-based stable signal to extract and identify, researchers focus on the transient signal. Today in the practical deployment of RFF, the most challenging is the solution for replacing high-end devices with low-end devices.

There is the theory of generating RFF. Transmitter imperfections are caused by the effect of tolerance from tolerance of electronics components, which generates RFF with analog elements, including phase noise, digital-to-analog converters, band-pass filters, frequency mixers, and power amplifiers [25], [31], [32] shows physical imperfections of transceivers elements with Fig. 3, including baseband filter, digital to analog converter, mixer, IF filter, amplifier, preselecting filter, analog to digital converter, digital signal processor,

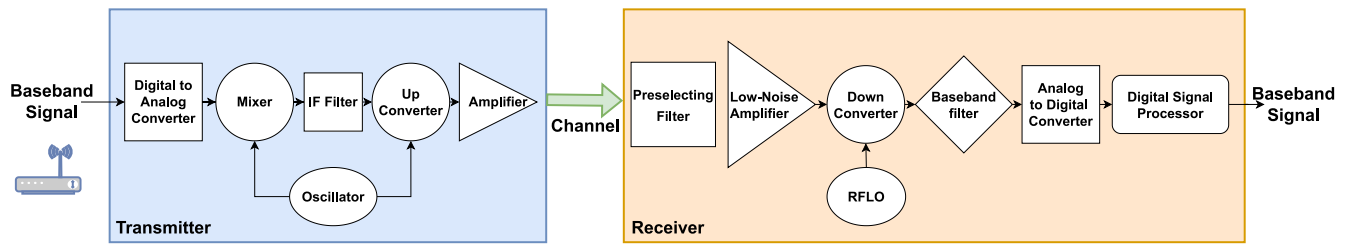


Fig. 3. Physical imperfections of transceivers elements [32].

etc. Although improving production accuracy could reduce the tolerance, the production cost would increase significantly, so the RFF generated from transmitter imperfections is available for most devices.

For achieving steady and high-accuracy identification, the property fingerprint should own the characteristics below [33].

- 1) Universality, which means that every wireless device should own the features that are able to be used for identification.
- 2) Uniqueness, which suggests that there are no both devices that own the same fingerprints.
- 3) Permanence, which indicates that the fingerprint should be time-invariant and environment invariant.
- 4) Collectability, which represents that fingerprint should be measurable and quantifiable, and collected with existing equipment.
- 5) Robustness, which expresses that the fingerprints should be evaluated steadily with influences on external environmental aspects like signal reflection, absorption, etc., and device-related aspects like temperature, power, and voltage level.

RFF identification is a method to identify wireless devices with RFF which generates from unique features of the wireless devices [34], [35]. In the past few years, commercial field utilizes many RFF methods [36]. For instance, RFF techniques used to identify and classify aircraft in ADS-B system belong to air traffic control [37]. According to the study of the RFF, it is feasible to implement authentication action.

In addition, [38] shows the physical-layer identification for Wi-Fi devices, including WRD-5620, WDR-5660, CPE-500, WS-5100, and MI-3A. It takes signal acquiring with coarse synchronization and fine synchronization. Then, it proposed two methods of RFF identification that Differential Phase of Pilots (DPoP) based on single dimensional features and Amplitude of Quotient (AoQ) based on long training sequences (LTSs). Finally, it got high identification accuracy with AoQ that 100% accuracy for WDR-5620, 100% accuracy for WDR-5660, 100% accuracy for CPE-500, 100% accuracy for WS-5100, 99% accuracy for MI-3A, and 99.3% for mixed NICs.

According to the previous study of identification with RFF, it got high accuracy of identification. It offers enough usability and reliability as a prepositive authentication or second authentication. Meanwhile, the accuracy of RFF identification will increase in the future with further study.

Besides, RFF authentication owns some security features to implement zero trust and secure access layer devices compared

with classical authentication, such as password authentication, including passwordless, universality, uniqueness, and robustness. Dissertation [39] studies the benefits and drawbacks for different stakeholders in password-based authentication.

In addition, classical authentication uses the fall back authentication to deal with the passwords lost, so the fall back authentication is still suitable for the RFF authentication, because RFF authentication owns the possibility of lost. However, the lost of RFF contains various reasons, such as device lost, device's RFF change, and identification error. We are able to set the short message service (SMS) authentication as fall back authentication.

III. RFF-BASED ZERO TRUST MECHANISM

A large scale network at an enterprise or organization would be divided into the core layer, distribution layer, and access layer in Fig. 4. The method proposed utilizes the RFF authentication to replace the password authentication at the access layer. Therefore, the mechanism avoids the weakness of password authentication on the AP and ensures a ZTA, including removal of center, resist data leakage, resist rogue AP attack, and security policy.

In Fig. 4, the solid red line with arrow means attacks to devices, and the arrow points to the attack direction. The dotted line with arrow means RFF authentication between devices, and the arrow points to the authenticators. The solid line with arrow means password authentication between devices, and the arrow points to the authenticators. The solid red line means attackers set up a rogue AP and connect to it. The question mark represents rogue AP attack aims to password authentication but RFF authentication and the exclamation mark means AP installed RFF authentication can identify the malicious devices with RFF and initiate alert.

As illustrated in Fig. 4, the common entry points of cyber attacks are devices at the access layer, because it is closest to the users or attackers, and it is lack of security devices due to closing to the security boundary or out of security boundary. Therefore, there are many traditional attacks on the devices at the access layer, such as spoofing identity, rogue AP attack, malicious terminals, sniffing attack, and password leakage.

Meanwhile, the password authentication and traditional security architecture cannot effectively eliminate the mentioned attacks. Besides, existing mechanisms rely on communication to transfer sensitive information, and sensitive information can be used to launch attacks. For handling

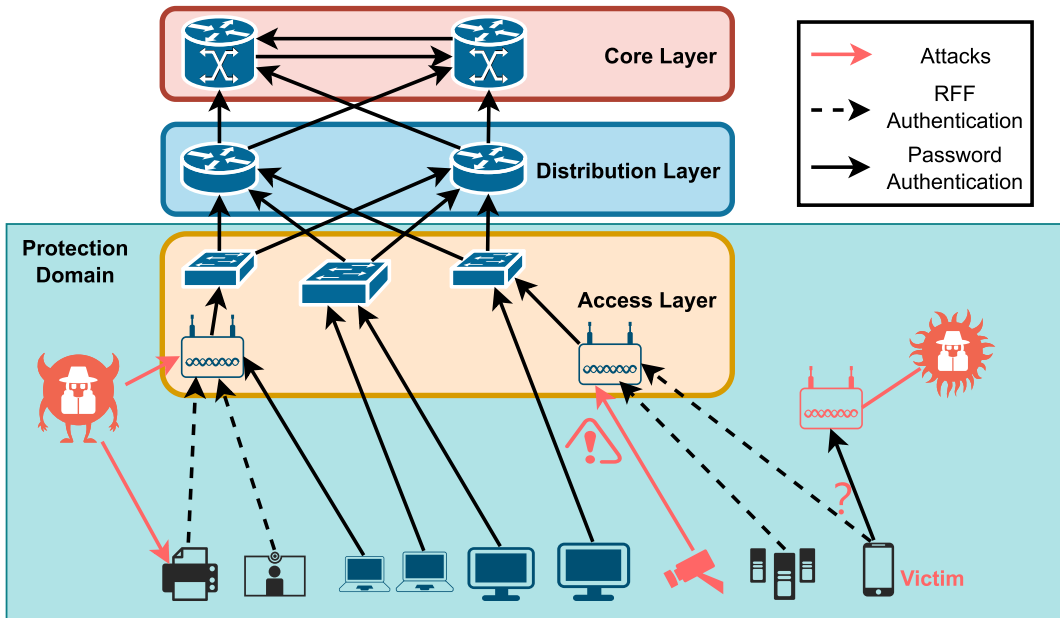


Fig. 4. Network scenario.

these problems, we design a resist data leakage function in Section III-B resist data leakage.

The rogue AP attack aims to get victim's sensitive information, such as password and the existing mechanisms are hard to identify rogue AP. Meanwhile, attacker can modify IP address and MAC address to avoid malicious devices detection. In addition, password leakage is caused by traffic capture and social engineering attack, and the existing mechanisms cannot stop social engineering attack and traffic capture. Section III-C resist rogue AP attack presents a function that resists rogue AP attack, and RFF resists password leakage with its passwordless character and security policy in Section III-D security policy.

With the zero trust and RFF techniques, these entry points will be protected, because the zero trust can decentralize risk within the systems. Therefore, not only the security center offers authentication function and high security strength but also the AP can complete this task and offers high security strength with support of RFF techniques. The details will be discussed in Section III-A removal of center.

As demonstrated in Fig. 5, the mechanism requires users to authenticate with AP, Server A, and Server B for getting different level privileges with different authentication methods. Besides, the RFF authentication is the first authentication, which makes the frequency of RFF authentication keep high frequency and lower authentication frequency of others. In addition, if RFF is lost, the user is able to enable fall back authentication to update RFF with SMS.

In our proposed mechanism, RFF authentication is applied at the access layer to offer security and adapts to ever-expanding or changing networks, namely, obscure or borderless networks, which is presented in Section II. Therefore, our mechanism makes protection at the access layer still take effect wherever devices are in the security boundary or not. Meanwhile, our mechanism overcomes the drawback

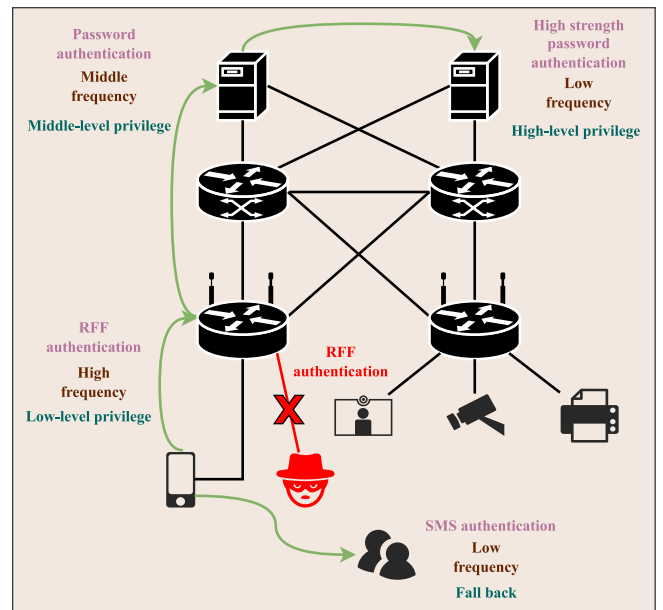


Fig. 5. Proposed mechanism.

of the classical authentication that it is weak between AP and terminal devices due to the performance limitation of AP.

According to the analysis of IoT wireless security problems in Section II and analysis above, the proposed mechanism should deal with the issues raised. Our mechanism would combine of zero trust and RFF techniques to handle these difficulties in the traditional security architecture. Therefore, we design and establish the following functions in our mechanism that removal of center, resist data leakage, resist rogue AP attack, and security policy.

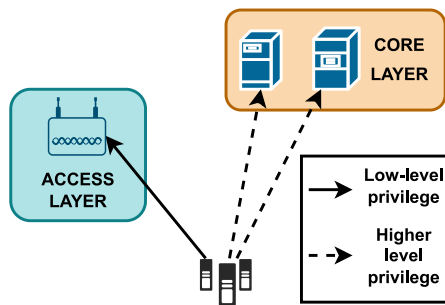


Fig. 6. Multiple authenticators in different network locations.

A. Removal of Center

With the zero trust concept, the access layer, distribution layer, and core layer on the network are trustless to each other, but the traditional security architecture relies on the reliability and reputation of the core layer. For example, when a new user visits the network, it should take authentication and authorization with the security service server on the core layer. Therefore, this traditional method cannot get rid of the dependence of security center. Meanwhile, if architecture puts the authentication to the access layer for removing the reliability of the security center, it would rise potential security concerns, because the password authentication on the AP owns weakness, especially the vulnerability. Equipment producers tend to focus on ensuring the security of core layer and distribution layer devices, because the traditional security architecture is based on a believable security center, so the likelihood of vulnerability appearing on access layer devices is higher than the likelihood of vulnerability appearing on core layer devices and distribution layer devices. Therefore, it needs a solid authentication method to ensure that an authentication implemented at the access layer is secure, which is RFF authentication. Consequently, taking RFF authentication is able to ensure secure authentication at the access layer without a security center. Meanwhile, it would accord with the ZTA concept, “default untrusted,” that the access layer, distribution layer, and core layer on the network are trustless to each other.

In other words, it relies on a secure element to establish a trust chain in traditional architecture. Therefore, the center provides a guarantee for each elements to build the expected environment, but the security is based on the center and it is easy to break the trust chain.

The removal of the center means multiple authenticators in different network locations like Fig. 6. In Fig. 6, the user should take authentication with an authenticator at the access layer for the low-level privilege, and the user should take authentication with different authenticators at the core layer for the higher privilege. Therefore, if any device at the access layer is compromised, the risk cannot expand to the core layer, namely, risk isolation.

B. Resist Data Leakage

With the zero trust concept, the proposed method takes RFF authentication at the access layer, so there is no password would transfer between the core layer and the access layer

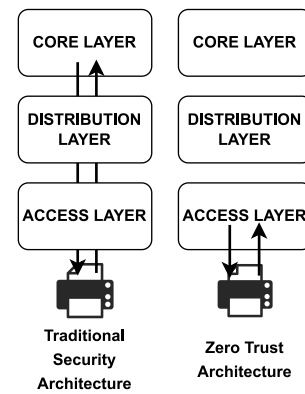


Fig. 7. Data flow between ITM and ZTA.

compared with traditional security architecture. The traditional security architecture needs users to access the wireless router, and offer an user ID and password to the authentication server at the core layer. During the flow of that mentioned, if hackers sniff the packets on the network, or each layer is compromised, which leads to the user ID and password leakage.

With the zero trust concept, the proposed method does not require the user ID and password stored in the authentication server at the core layer, so if the authentication server is compromised, the sensitive information, user ID, and password, would not leak and infect other accounts’ security on other systems. Therefore, the influence of compromising the authentication server cannot spread to other network layer devices, namely, risk isolation.

If the RFF authentication is invalid due to the device lost or RFF change, it requires the administrator to enable the fall back authentication with SMS for resetting the invalid RFF, which enhances the security by security management. Even if the staff is compromised and the staff makes the RFF authentication compromised, this risk cannot cause password exposure at the core layer.

As illustrated in Fig. 7, the traditional security architecture authenticator is not at the access layer, so the authentication information will through the access layer, the distribution layer, and the core layer, which suffers the potential risk of data leakage when any device is compromised at any network layer. However, ZTA supports authentication with the device at the access layer, because RFF identification ensures the security of authentication compared with the classical authentication method. In ZTA, the authentication information does not through all the network layers, which reduces the risk of data leakage with devices compromised at the distribution layer and the core layer. Meanwhile, the RFF authentication is applied to most scenarios, so password authentication for getting higher privilege seldom appear on the network.

C. Resist Rogue AP Attack

Rogue AP attack is aim at getting sensitive information, such as user name, password, and token. However, the RFF authentication could prevent leakage of Wi-Fi passwords and spoof attacks compared with password authentication, because the RFF authentication uses the RFF to authenticate with AP,

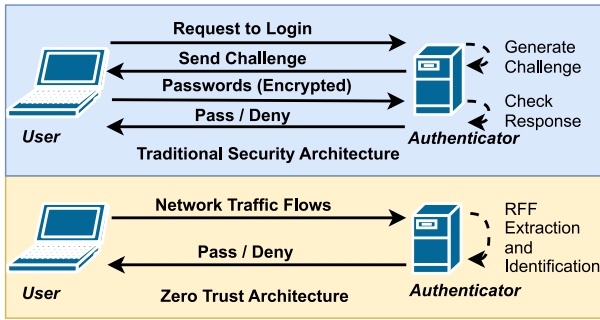


Fig. 8. Authentication between password authentication and RFF authentication.

and the RFF owns the no-cloning character. According to the no-cloning character, the attacker sniff cannot clone the RFF of the legal device, because the attacker's device owns RFF features on the transmitter and receiver which cannot be removed, which is demonstrated in Section IV-A defensive with no-cloning of RFF.

Though two-way authentication is a feasible solution for improving password authentication, which solves the spoof attack. However, two-way authentication consumes large network and computational resources, which makes the attacker take the Denial of Service (DoS) attack easily, so there is not a great solution for dealing with rogue AP attacks, and RFF authentication could handle rogue AP attacks.

The RFF authentication offers another authentication method, which conforms to the zero trust concept that always verifies. The different authentication reduces the risk due to low likelihood of appearing common vulnerabilities, for example, the rogue attack for password authentication makes spoofing attack feasible, but the RFF authentication does not own this risk. Therefore, setting the RFF authentication as prepositive authentication and setting password authentication as postpositive authentication is a feasible solution.

As illustrated in Fig. 8, in traditional security architecture, users transmit encrypted passwords to an authenticator, and it is possible that the attacker stores connected users' passwords with the rogue AP attack. Nevertheless, in the ZTA, the user just sends RFF to the authenticator, so attackers only gain the RFF with the rogue AP attack. Therefore, based on the no-cloning feature of RFF, the attacker cannot clone the legal RFF gained with the rogue AP attack.

D. Security Policy

The architecture still needs some common protection methods, such as network isolation according to workspace, access control system, and other common security policies. There is a security policy about RFF authentication with zero trust concept, "always verify," that user who completes the RFF authentication would get low-level permissions, the user who completes the authentication with the security center of device A at the core layer would get middle-level permissions, the user who gets authorization with the security center of device B at the core layer would get high-level permissions temporary, as Fig. 9 shows.

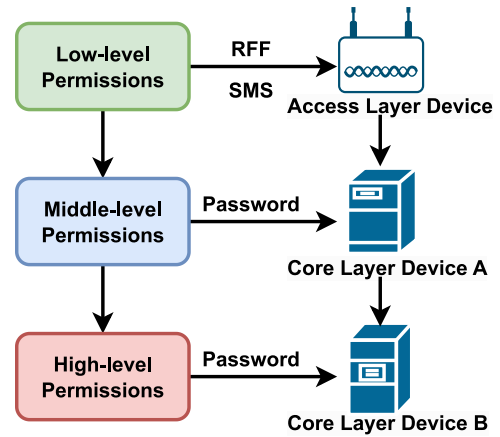


Fig. 9. Security policy.

As Fig. 9 demonstrated, the attackers want to gain low-level permissions means the access layer devices are compromised, and they gain middle-level permissions with the core layer device A compromised and gain high-level permissions with the core layer device B compromised, which exists risk diversification. Meanwhile, the security policy exists ordinal relation, which means if the attacker cannot get the low-level permission by cracking RFF authentication, they cannot get the higher permissions, so the security of RFF authentication ensures the security of the authentication system. In addition, if the RFF authentication becomes invalid with the device lost and RFF change, the user should contact the administrator to enable fall back authentication with SMS authentication for resetting the invalid RFF.

IV. SECURITY ANALYSIS

This section presents the security analysis of the proposed method, including defensive with no-cloning of RFF, RFF authentication for improving security, and threat model. Defensive with no-cloning of RFF proves the RFF authentication cannot clone by attackers, which means if attackers collect the RFF, the attacker still cannot copy the legal RFF. Besides, RFF authentication for improving security demonstrates that RFF authentication makes secure authentication at the access layer become feasible in the zero trust security architecture. Meanwhile, we use the STRIDE threat model to evaluate the security of the proposed mechanism and present advantages.

A. Defensive With No-Cloning of RFF

The AP device would offer the RFF identification for users, the identification mechanism is using machine learning to train a model to identify the wireless device want to access, and finally, the AP would own a model to match the signal from the wireless device with different features on different weights. Therefore, it avoids security concerns of WEP, WPA, and WPA2, that the password could leak, recover, and skip. In the RFF identification mechanism, the attacker should simulate the signal for the illegal wireless device, which would take massive cost, because attackers need a high accurate equipment to capture the signal from legal wireless devices,

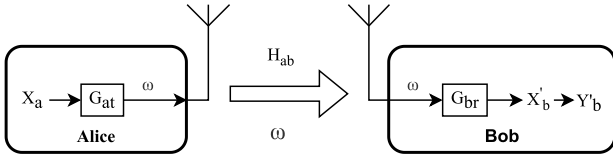


Fig. 10. Normal generating RFF.

and it is hard to get which features taking the weight when RFF identification processing and the high accurate equipment simulating signal would take large cost.

The reason why RFF identification is hard to crack is that it is infeasible for attackers to clone RFF from legal devices. The normal flow of generating is illustrated in Fig. 10, Alice sends a signal (X_a) to Bob, which is affected by RFF features of Alice's transmitter (G_{at}), channel (H_{ab}), noise (ω), and RFF features of Bob's receiver (G_{br}), then Bob generates the RFF of Alice (Y_b) on Bob's device when Bob received the signal (X_a).

The flow of cloning RFF demonstrates in Fig. 11, Alice sends Signal (X_a) to Bob, but it is captured by attacker Eve, the Signal received by Eve (X_e) is affected by RFF features of Alice's transmitter (G_{at}), channel (H_{ae}), noise (ω_2), and RFF features of Eve's receiver (G_{er}). Then, the attacker Eve sends signal (X_e) with cloning signal (X_a) to Bob, which is affected by RFF features of Eve's transmitter (G_{et}), channel (H_{eb}), noise (ω_1), and RFF features of Bob's receiver (G_{br}). Finally, Bob generates the counterfeit RFF of Alice (Y'_b). Due to the flow of cloning RFF, the RFF (Y_b) and counterfeit RFF (Y'_b) are highly different.

There are three formulas that present the generation of normal RFF (Y_b), the generation of counterfeit RFF (Y'_b), and the differences between RFF (Y_b) and counterfeit RFF (Y'_b), (1) and (2) show constitute of Y_b and Y'_b , respectively. Formula (3) demonstrates difference between RFF (Y_b) and counterfeit RFF (Y'_b).

In (1), X_a means signal sent by Alice, G_{at} means RFF features Alice's transmitter owned, G_{br} means RFF features Bob's receiver owned, H_{ab} means the effect of a channel between Alice and Bob, and ω means noise effect among Alice and Bob. The flow of generating RFF of Alice, namely Y_b , is Alice sending signals with the effect of RFF features Alice's transmitter owned, then it affected by the channel between Alice and Bob and noise, then signal received by Bob with the effect of RFF features Bob's receiver owned. The (1) is given as

$$Y_b = X_a \bullet G_{at} \bullet G_{br} \bullet H_{ab} + \omega. \quad (1)$$

In (2), X_e means signal sent by Eve, G_{et} means RFF features Eve's transmitter owned, G_{br} means RFF features Bob's receiver owned, H_{eb} means the effect of a channel between Eve and Bob, and ω_1 means noise effect among Eve and Bob. Formula (2) takes the decomposition of X_e , the factors are the same as the formula but the ω_1 , which is the noise among Alice and Eve. Besides, G_{et} means RFF features Eve's receiver owned, and H_{eb} means the effect of the channel

between Alice and Eve. The (2) is given as

$$\begin{aligned} Y'_b &= X_e \bullet G_{et} \bullet G_{br} \bullet H_{eb} + \omega_1 \\ &= (X_a \bullet G_{at} \bullet G_{er} \bullet H_{ae} + \omega_2) \bullet G_{et} \bullet G_{br} \bullet H_{eb} + \omega_1. \end{aligned} \quad (2)$$

In (3), it is the difference between RFF Y_b and counterfeit RFF Y'_b . X_a , G_{at} , and G_{br} are common factor of Y_b and Y'_b . It is noticeable that noise ω_1 and ω_2 make difference large. If there is no noise environment, the difference is affected by the effect of the channel between Alice and Eve (H_{ae}) and the channel between Eve and Bob (H_{eb}). if there are no noise and no channel influence, the difference is affected by the RFF features Eve's receiver owned (G_{er}) and the effect of RFF features Eve's transmitter owned (G_{et}). However, it cannot assume a situation that there is no effect of the device's RFF features, because it is an inherent attribute. The (3) is given as

$$\begin{aligned} \Delta(Y_b Y'_b) &= X_a \bullet G_{at} \bullet G_{br} (G_{er} \bullet H_{ae} \bullet G_{et} \bullet H_{eb} - H_{ab}) \\ &\quad + \omega_2 \bullet G_{et} \bullet G_{br} \bullet H_{eb} + \omega_1. \end{aligned} \quad (3)$$

Based on the above analysis, the RFF owns no-cloning character. Therefore, any access attacks in wireless network scenario, such as replay attack and spoofing attack cannot achieve with the copy of legal devices' signals.

B. RFF Authentication for Improving Security

RFF authentication handles several problems within the zero trust security architecture below. The RFF techniques utilize collections of signals from different wireless devices to identify legal devices. The RFF identification can be implemented by feature engineering and deep learning. The feature engineering method should extract the specific features from processed received signals with professional knowledge, and the deep learning method utilizes processed or raw collections of signals to train model for identification. There are points described as the benefits following.

- 1) RFF authentication resolves the rogue AP attack and malicious terminals within the access layer, because RFF cannot be copied perfectly due to the no-cloning feature of RFF. Therefore, if the attacker captured the RFF information from the signal, the attacker still cannot send a legal RFF to the authenticator. Meanwhile, the legal AP is able to find the malicious terminals, because the malicious terminals holds strange RFF. Rogue AP attack is a problem for password authentication, though two-way authentication is a solution, two-way authentication costs higher resources and low defense for DoS attacks, so RFF authentication is a good choice compared to password authentication.
- 2) RFF authentication makes prepositive authentication at the access layer feasible, meanwhile, the authentication information should not send to the authenticator at the core layer, so the risk of capturing authentication from other network locations will reduce compared to the password authentication. Besides, the RFF is essential authentication information for the user, but it is useless for the attacker, because RFF cannot be copied for illegal authentication. Compared to password authentication,

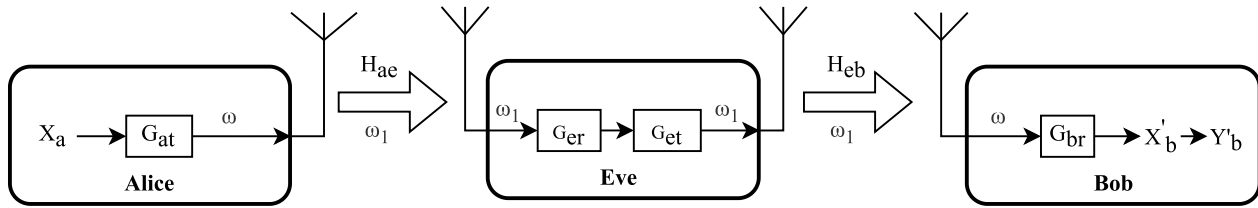


Fig. 11. Clone attack for RFF.

the password is the most important factor for authentication and the password is possible to leak, so RFF authentication improves the security of authentication.

Based on the advantages of the RFF authentication analysis above, RFF authentication is able to secure the authentication, resist attacks, and support the implementation of zero trust. RFF authentication owns the features that classical authentication does not have, such as uniqueness generated from physical phenomenon.

C. Threat Model

With the STRIDE threat model, the proposed method would be analyzed in terms of spoofing identity, tampering with data, repudiation threats, information disclosure, DoS, and elevation of privileges. We utilize the STRIDE threat model to analyze threats and potential threats of the proposed mechanism. Meanwhile, we can measure the robustness with various threats.

1) *Spoofing Identity*: The spoofing identity threats include signal counterfeit attack, identity impersonation attack, and signal replay attack. Signal counterfeit attack is forging a legal user's signal to get the authentication. When the attacker transmits the legal users' signal, the RFF generation on the receiver will be affected by the attacker's transmitter and the channel between the attacker's device and receiver, so the signal counterfeit attack is unable to crack RFF authentication.

Identity impersonation attack is assuming the attacker knows the authentication scheme and cracks it. However, the RFF authentication used in the method owns a noncloning feature, so the attacker cannot offer the legal identity information that legal RFF.

Signal replay attack [40] is attacker replays the legal users' signal to the receiver for getting the authentication. Nevertheless, the RFF generation on the receiver will be affected by the attacker's receiver, the attacker's transmitter, and the channel between the attacker's device and receiver, so the attacker cannot generate a legal RFF on the receiver.

2) *Tampering With Data*: The tampering threats include packet injection, spoofing attack, access to logs, and sensitive information. Packet injection is injecting code to run on the target server with packets. However, the proposed method owns two different authenticators that device at the access layer and security server at the core layer, so the packet injection should attack both authenticators for higher level permission or risk.

Spoofing attack is forging the packet to impersonate a legal user. However, the situation is the same as the packet injection mentioned, the two different authenticators and authentication

methods ask the attacker to take higher cost for attacking the authentication system. For example, if an attacker uses vulnerability to attack the authentication, the attacker needs one vulnerability for the access layer device and one vulnerability for the security center.

Access to logs and sensitive information is that the attacker aims to get the sensitive information on the authentication server, such as passwords. Nevertheless, there are two authenticators, so there are two databases stored the password, which means the attacker should access two databases. Besides, though the attacker knows the legal RFF, the attacker cannot clone the legal RFF from illegal devices for sending to the receiver due to the noncloning feature of RFF.

3) *Reputation*: Repudiation means a proof of bad actor indeed taking bad operations. The legal IP address and MAC address are feasible to forge, but the legal RFF is infeasible to forge. It is possible to change the RFF, but if the legal RFF changed, the user cannot recover the legal RFF, because the RFF is generated from the imperfection of hardware, so changing RFF means changing hardware, which is irreversible.

4) *Information Disclosure*: Information disclosure includes rogue AP attack, password leakage, and sniffing attack. Rogue AP attack means attackers set a rogue AP for attracting legal users to access it, leading to the attacker gaining information from legal users, such as passwords, messages, and documents. This method does not need a password to complete the prepositive authentication, so the sensitive information leakage cannot make attackers complete the prepositive authentication, namely RFF authentication. The RFF authentication requires the user to generate a legal RFF with signal, but the noncloning feature of RFF makes the attacker unable to simulate legal users.

Password leakage means the password is disclosed by users' behavior or the password is collected by a hacker with various approaches [41], [42]. With the first method to gain users' passwords, based on RFF authentication users need not write a password or store a password to prevent from forgetting password with memory burden of users. With the second method to achieve users' password, the attacker cannot feign the legal user by information collected with the noncloning feature of RFF.

Sniffing attack means the attacker sniffs packets over the network and gets sensitive information. In the method, there is no password transmitting over the network, because the prepositive authentication is RFF authentication with the access layer device. However, in the classical method users will authenticate with the authentication center, which makes password transmitting over the network easier to be taken sniffing attack.

5) *Denial of Service*: The DoS means attackers occur the resources of the target to break down devices, networks, or services with various methods, such as flooding. The DoS could send continuous RFF authentication requirements to the access layer device, which is still useful for the RFF authentication-based method. Besides, with the IoT devices' limitation of energy, the DoS attack includes energy harvesting by sending a large number of legitimate or corrupted messages.

6) *Elevation of Privilege*: Elevation of privilege means the attacker is able to access information or services without proper privileges. The attacker cannot complete the RFF authentication as prepositive authentication due to the non-cloning feature of RFF, but the attacker is able to complete middle-level permission and high-level permission authentication with sniffing and spoofing attacks. However, most authentications are completed by RFF authentication, so the occurrence number of passwords will greatly decrease over the network, contributing to the risk of credentials leakage with sniffing and spoofing attack reduction.

In summary, the proposed mechanism owns high robustness in spoofing identity, tampering with data, and information disclosure. However, it is weak in reputation and it cannot against DoS. In terms of elevation of privilege, our mechanism decreases the risk by robustness in information leakage, but it cannot defend against vulnerabilities.

V. PERFORMANCE EVALUATION

In addition to the security evaluation, the performance evaluation is essential, including accuracy experiments, cost experiments, and collision experiments. The accuracy experiments and collision experiments represent stability, namely, whether the normal requirements are met with the identification accuracy to identify legal devices and users. The cost experiments represent whether the authentication is suitable for various scenarios and users, including small businesses, large enterprises, etc.

The RFF authentication performance evaluation takes 54 TI CC2530 ZigBee modules. Besides, the receiver to collect ZigBee RF waveforms is universal software radio peripheral (USRP), and the PC installs 11th Gen Intel Core i7-11700K without a graphic card. The experimental setup is demonstrated in Fig. 12. The raw data is generated from the 54 ZigBee devices and the PC takes raw data processing, training, and classification offline.

The RFF authentication is replaceable, so the RFF identification can be replaced by RFF identification methods from other researchers according to requirements and scenarios, and we evaluate two proposed RFF identifications that hybrid RFF extraction and device classification scheme [33], and deep learning-based RFF identification using differential constellation trace Fig. [43] for proving feasibility of RFF authentication.

Hybrid RFF extraction and device classification scheme utilize four modulation features, including differential constellation trace figure (DCTF), frequency offset, I/Q offset, and circle offset. This method is based on feature engineering, and the flow is demonstrated in Fig. 13 [33]. This scheme extracts



Fig. 12. Experiment setup for the RFF feature extraction.

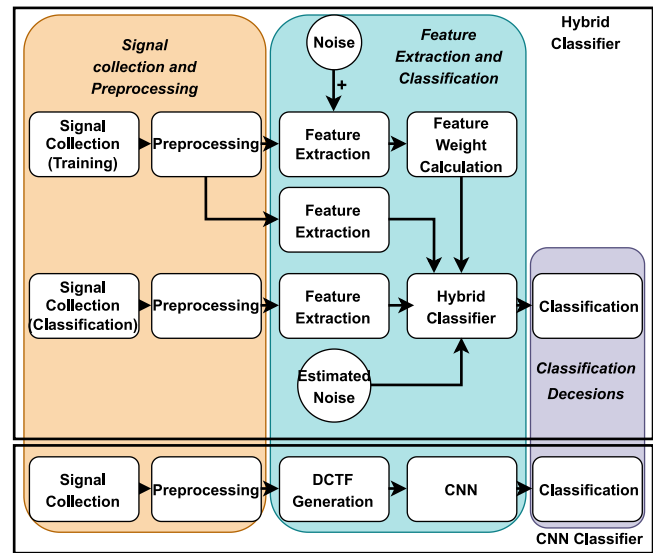


Fig. 13. Hybrid classifier and CNN classifier [33], [43].

feature with high signal noise ratio (SNR), and save these feature as the template in the training stage. Then, it adds additive white Gaussian noise (AWGN) to the capture signal, and calculate different feature weight for the training model to against SNRs. In the classification stage, it will take feature extraction and SNR estimation, and find the most appropriate feature weights with estimation. Then, it utilizes the features extracted and the feature weight to classify the RFF.

Deep learning-based RFF identification utilizes convolutional neural network (CNN) and DCTF. The flow of this method is illustrated with Fig. 13 [43]. It generates DCTF with each device, and each DCTF is the fingerprint for each device. Then, the CNN model trains and classifies DCTFs for identification. The CNN consists of four layers that three convolutional layers and one fully connected layer.

According to the accuracy experiments, collision experiments, and cost experiments below, we find that both RFF identification methods are low cost and robust in collision, but they own different advantages in accuracy and time cost. In this article, we do not focus on feature engineering or

TABLE II
PERFORMANCE OF DIFFERENT METHODS

Method	Accuracy	Training Time	Recognition Time
Hybrid RFF extraction and device classification scheme	95.10%	9.87 seconds	0.14 seconds
Deep learning based RFF identification using differential constellation trace figure	99.10%	142 seconds	2.09 seconds

TABLE III
ACCURACY OF HYBRID RFF EXTRACTION AND DEVICE WITH DIFFERENT SNRS

SNRs in Training \ SNRs in Test	5 dB	10 dB	15 dB	20 dB	25 dB	30 dB
5 dB	92.51%	86.47%	69.05%	56.31%	33.46%	46.54%
10 dB	92.11%	95.28%	91.55%	91.27%	89.75%	89.27%
15 dB	92.01%	94.79%	93.88%	92.73%	92.26%	90.52%
20 dB	92.13%	93.88%	93.48%	94.91%	93.06%	93.11%
25 dB	93.18%	93.66%	92.16%	93.74%	97.49%	93.59%
30 dB	93.41%	93.55%	92.01%	93.01%	92.71%	94.18%

deep learning methods. For instance, 5G defines three major application scenarios, including enhance mobile broadband (eMBB), massive machine type communication (mMTC), and ultra reliable and low latency communication (uRLLC). uRLLC focuses on low latency of access devices, while mMTC focuses on access of massive devices. Therefore, feature engineering-based method offers lower training time and recognition time, which is suitable for uRLLC scenarios. Deep learning-based method offers higher accuracy, which is suitable for mMTC scenarios, and higher accuracy means it can be applied to larger-scale IoT terminal access scenarios. Meanwhile, the training time, recognition time, and accuracy are demonstrated in Table VI.

A. Accuracy Experiments

There is a comparison between both methods with the same raw data, including 54 devices, we compare the accuracy and time with Table II. The hybrid RFF extraction and device classification scheme experiment is performed with 95.10% accuracy. Besides, the training time is 9.87 s, and the recognition time is 0.14 s. With the above results, it is proved that low cost and high accuracy are possible and feasible.

The deep learning-based RFF identification using DCTF is trained five times, and verified three times after each training. The experiments prove it keeps 99.10% accuracy, though it takes 142 s for training. However, the recognition time is acceptable, because it is 2.09 s. This method sacrifice cost to improve accuracy, which means higher accuracy is possible and feasible compared with the hybrid classifier-based method mentioned.

Comparison between two methods indicates the RFF identification-based CNN is better in terms of accuracy, but the

RFF identification-based hybrid classifier owns a lower cost. Besides, there are many RFF identification methods proposed by researchers that can be used for RFF authentication, users need to consider the contexts and requirements to decide an applicable method.

In addition, the robust performance related experiment is accomplished, the experiments test the accuracy between two methods with different SNRs in the test scenario and training scenario, and we demonstrate results with Tables III and IV, respectively. Meanwhile, considering the collected data exist a level of noise, these experiments started SNR in the test and SNR in the training from 5 dB.

According to Tables III and IV, the hybrid classifier-based method owns higher robustness when devices are under low SNR scenarios. Therefore, the RFF authentication is possible and feasible to possess high robustness.

B. Collision Experiments

In addition, there are experiments with brute-force attacks to test the collision of the ZigBee devices' RFF and the defense of passwords authentication with a weak password, because RFF authentication does not exist passwords, so attackers need to utilize the collision of the ZigBee devices' RFF to crack the devices. We utilize the Python program to test the time cost of cracking weak passwords with dictionary attacks, the average time cost is 0.214 s with ten times tests.

Besides, the experiments related to the collision of the ZigBee devices' RFF use 54 devices, and taken three experiments, and all experiments apply the hybrid RFF extraction and device classification scheme. Besides, the scheme takes the features' distance to calculate the distances between RFFs, including distances of DCTF, distances of frequency offset,

TABLE IV
ACCURACY OF DEEP LEARNING-BASED RFF IDENTIFICATION WITH DIFFERENT SNRS

SNRs in Training \ SNRs in Test	5 dB	10 dB	15 dB	20 dB	25 dB	30 dB
5 dB	1.85%	1.85%	3.78%	16.41%	62.26%	99.1%
10 dB	1.85%	1.85%	3.75%	16.44%	62.45%	99.13%
15 dB	1.85%	1.85%	3.75%	16.31%	62.3%	99.14%
20 dB	1.85%	1.85%	3.8%	16.37%	62.05%	99.03%
25 dB	1.85%	1.85%	3.77%	16.35%	62.38%	99.05%
30 dB	1.85%	1.86%	3.77%	16.37%	62.35%	99.09%

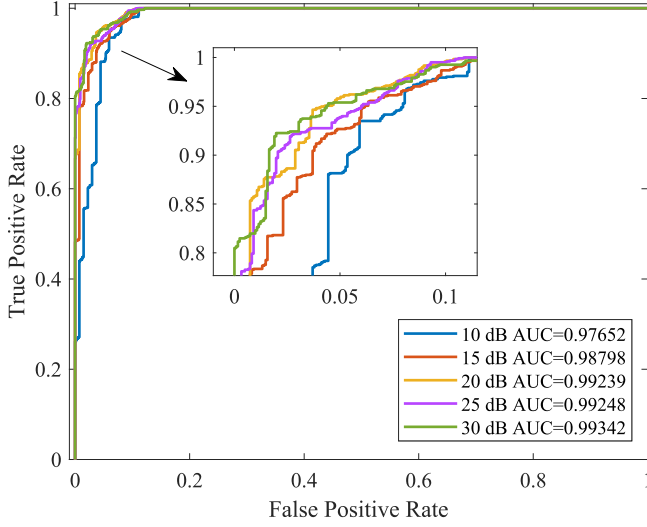


Fig. 14. ROC curves for hybrid RFF extraction and device classification scheme with 27 legal devices under different SNR.

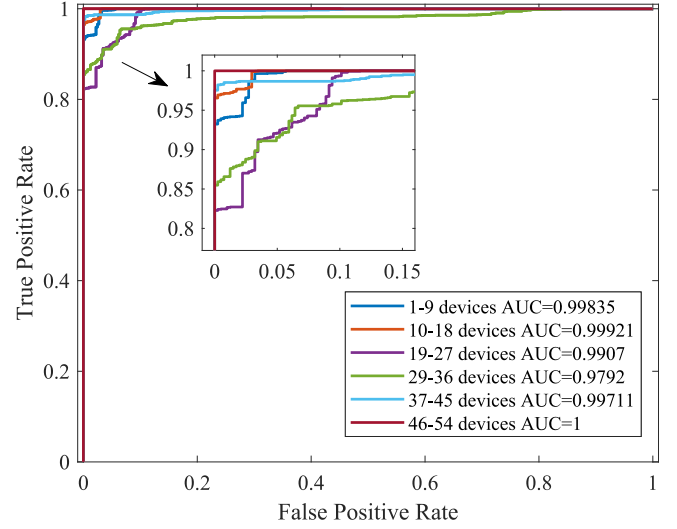


Fig. 15. ROC curves for hybrid RFF extraction and device classification scheme with different groups of nine legal devices.

distances of I/Q offset, and distances of circle offset [33]. The distance between devices' RFF is defined as

$$D = \omega_1 \cdot d_1 + \omega_2 \cdot d_2 + \omega_3 \cdot d_3 + \omega_4 \cdot d_4 \quad (4)$$

where D is the distance between devices' RFF, the $\omega_1, \omega_2, \omega_3$, and ω_4 are the weights of the each feature's distance, the d_1 is the distance of DCTF, the d_2 is the distances of frequency offset, the d_3 is the distances of I/Q offset, and the d_4 is the distances of circle offset.

The first experiment uses 27 devices as legal devices and 27 devices as illegal devices, we find that the distances between legal devices' RFF and the legal RFF in the database are lower than 0.067122, and ROC Curve is demonstrated with Fig. 14. The second experiment divides 54 devices into 6 groups with 9 devices in each group, and it sets one group of devices as legal devices at a time, and then calculates the RFF distances, we find that the distances between legal devices' RFF and the legal RFF in the database are lower than 0.007915, and ROC Curve is demonstrated in Fig. 15. According to these experiments, it is proven that the scheme is robust to deal with the collision of the ZigBee devices' RFF.

C. Cost Experiments

1) *Time Cost*: We take experiments to evaluate the time cost between classical identity authentication and RFF identity authentication. The experiment setup includes one PC installed 11th Gen Intel Core i7-11700K without a graphic card, one laptop installed Intel Core i5-5257U with Intel Iris 6100 graphics, and a router to connect many PCs, including the two laptops, which is in an office network. We find that the AES encryption and decryption are fast based on the PC, and the time cost is 0.0004 s. However, users should transfer the passwords to the trustable center in classical authentication, and the transmission time is around 0.032 s in a network situation without traffic busy and around 5.08 s in a network situation with traffic busy. Compared with the classical method, the RFF identification just needs to communicate with the device in the access layer and takes 0.008 s, which is caused by the removal of the center described in Section III-A removal of center.

The processing time of RFF identification is 0.12 s based on the hybrid RFF extraction and device classification scheme, and the processing time of feature extraction and model training is 9.87 s. Nevertheless, the processing time of

TABLE V
TRAFFIC BUSY SITUATION AND CORRESPONDING TRANSMISSION TIME

Index	Packet Sending Rate	One Time Transmission Time	Challenge-response Authentication Mechanism Transmission Time
1	0 packet/second	0.008 seconds	0.032 seconds
4	1000 packets/second	0.009 seconds	0.036 seconds
5	10000 packets/second	0.032 seconds	0.128 seconds
6	12000 packets/second	0.899 seconds	3.596 seconds
7	14000 packets/second	1.073 seconds	4.292 seconds
8	16000 packets/second	0.821 seconds	3.284 seconds
9	18000 packets/second	0.985 seconds	3.940 seconds
10	20000 packets/second	1.535 seconds	6.140 seconds

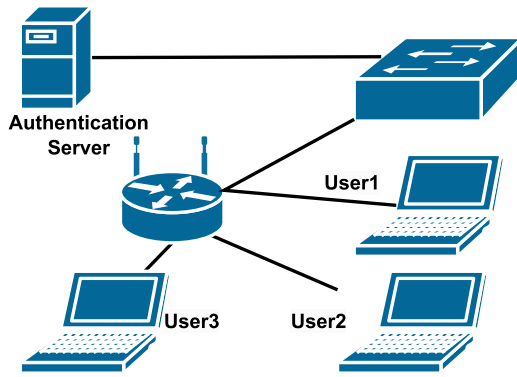


Fig. 16. Time cost experiment network topology in reality.

feature extraction and model training can be further compressed in practical application systems, because many feature parameters need to be estimated and compensated in the communication process, such as carrier frequency offset. Therefore, the majority time cost is the processing time of RFF identification for AP devices containing RFF authentication function. Meanwhile, according to these experiments, it is proven that RFF authentication is slower than the classical authentication method in the scenario without busy traffic, but it is faster in the traffic busy scenario. The time cost between the two methods demonstrates in Table V.

The RFF authentication does not send packets through the whole network and does not follow the challenge-response authentication mechanism in Fig 8. The RFF authentication only requires the user to authenticate with AP with two interactions and it is demonstrated in Fig 8. Therefore, the transmission time in reality is 0.008 s. With the challenge-response authentication mechanism demonstrated in Fig 8, the transmission time without traffic busy is 0.032 s and the transmission time with traffic busy is 5.08 s.

We take the experiment in a real situation with real traffic in the network, we measure the transmission time is around 0.008 s. The topology is demonstrated in Fig. 16 below. In addition, we use the Hping3 tool to send the packet with Users 2 and 3 to the server for simulating traffic busy, then we use the laptop called User 1 to send the packet to the

TABLE VI
TIME COST BETWEEN AES-BASED METHOD AND RFF-BASED METHOD

Methods	Transmission Time	Process Time	Total Time
Method based AES without traffic busy	0.032 seconds	0.0004 seconds	0.0324 seconds
Method based AES with traffic busy	5.08 seconds	0.0004 seconds	5.0804 seconds
Method based RFF	0.008 seconds	0.1200 seconds	0.1280 seconds

server and measure the transmission time. In the traffic busy situation, we measured transmission time with three results that are around 0.008 s, various times within 1 s such as 0.71 s, and more than 1 s such as 1.27 s. These, respectively, represent that the packet is not in the queue by chance, the packet is in the queue, and the packet is in the queue but timeout. The reason why transmission times are changing, is that the traffic does not stay at a consistently high level, which is fluctuant. The experiment result of traffic busy situations and the corresponding transmission time is illustrated in Table VI.

We found that when the packet sending rate is 10 000 packets per second, the transmission time of classical authentication is more than the total time of RFF authentication. The 10 000 packets per second means 1.659 MB/s, because the length of each packet we sent is 174 Byte. For instance, when we start a video online, the network speed keeps 3.75 to 6.25 MB/s at a short moment of the beginning. Therefore, if large-scale authentication takes at the same time within a system, it will cause network latency.

2) *Storage Cost*: Before we run the RFF identification system, we measure the storage of data that should be used when we are in the deployment and setup of the RFF identification phase. The storage costs are mainly focused on the training phase, because the model will read frames in sequence from the received signal and each frame read means one-time identification, which does not have a storage requirement. Meanwhile, one collected signal occupies around 5 MB, including 18 frames, so one frame only occupies around 0.278 MB. Besides, we utilize 54 ZigBee devices to train the model, so the storage of all signals used is 263 MB. Each device that needs to be identified by the system will occupy around 5 MB of storage space.

After we run the RFF identification system, we measure the storage of the trained model. The method based on feature engineering that Hybrid RFF extraction and device classification scheme should store the extracted features, which occupies 10 KB. The method based on deep learning that Deep learning-based RFF identification using DCTF should store neural network model, which occupies 3.21 MB. In summary, both systems have low storage requirements.

3) *Algorithm Cost*: The cost of the algorithm is gradually decreasing with the development of the market and technology. Some feature extraction algorithms in feature engineering methods can be integrated into the baseband chip, such as the extraction of CFO features. In addition, another algorithm can also be added to the design of the chips by field programmable gate array (FPGA) after these algorithms become mature.

Deep learning algorithm requires extra costs, such as GPU. However, some routers and many embedded systems are integrated with GPU-accelerated designs, which is satisfied with deep learning algorithm cost. For example, the Nanopi R5C Mini WiFi Router integrated with a GPU module called Mali-G52 1-Core-2E and R86S-T3 industrial router integrated with Intel UHD graphics. Besides, an embedded system called Raspberry Pi 5 owns a GPU module, VideoCore VII GPU. Therefore, RFF-based authentication mechanism will be feasible with the low cost GPU-accelerated devices.

VI. CONCLUSION

In this article, an authentication method based on zero trust with RFF is proposed. With the study of the zero trust, we deploy the principles of zero trust and the core idea, never trust, always verify, to the authentication method. Compared with classical password authentication, we utilize the RFF authentication, because we attempt to remove the center for the “never trust” idea. Therefore, the proposed method does not always rely on the trust-ability of the authentication center or security center. The prepositive authentication authenticates with the access layer device, and the RFF authentication offers higher security compared with classical passwords due to the differences in the authentication and noncloning feature of RFF, which ensure the authentication taken at the access layer is still secure.

According to the analysis in this article, the mechanism has the ability to resist data leakage and resisting rogue AP attack. Meanwhile, it follows the core idea of zero trust that “always verify,” then it decreases the risk of password exposure with multiple authentications, including RFF authentication. Besides, with the STRIDE threat model analysis, the proposed method owns high-level security, especially information disclosure and spoofing identity.

Related to the performance, the accuracy keeps 99% and the time cost is 142 s, which can be applied in most scenes with the performance. Besides, the accuracy and cost could be increased in the future, because the identification method is a replaceable component, it is a good choice to use better identification methods from other researchers with detailed requirements and contexts. Moreover, if the mechanism deploys the hybrid-based RFF authentication, the 95% accuracy is acceptable and the robustness is high. In addition, according to experiments RFF authentication owns the ability of fast authentication and robustness against collision.

In terms of implementation suggestions, we demonstrate the electric system scenario as an instance. There are massive sensors constantly uploading the monitoring information to the center for distributing power efficiently. Therefore, Based on infrastructure security considerations, if traditional authentication methods are used for every data upload, it will cause huge costs. Nevertheless, Using our mechanism with RFF authentication for every communication can effectively reduce costs. Besides, the Vehicle to Vehicle (V2V) scenario in the Internet of Vehicles (IoV) scenario involves identity impersonation attacks, which can spread false information or take DoS. However, if it uses RFF-based authentication

mechanism, with the noncloning feature of RFF, the malicious vehicle cannot impersonate other vehicles.

ZTA combined with the RFF authentication securely ensures multiple different authentication methods and multiple different authenticators for reducing risk. Besides, RFF authentication reduces the risk when it is authenticated with an access layer device compared with password authentication. RFF authentication supports ZTA in this method to complete “no zero, always trust” for ensuring constant control and risk reduction.

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